

Tarea №3

(**Deadline:** May 27, 2026)

1. (**4 points**) A relativistic particle of mass m interacts with an external symmetric rank-2 tensor field $S^{\mu\nu}$, so the action is given by

$$S = \int d\tau \Lambda, \quad \Lambda = -mc^2 \sqrt{\dot{x}_\mu \dot{x}^\mu} + e \dot{x}_\mu \dot{x}_\nu S^{\mu\nu}(x) \quad (1)$$

where τ is the proper time, $x^\mu(\tau)$ is the trajectory of the particle, and $\dot{x}^\mu = dx^\mu/d\tau \equiv u^\mu$ is the 4-velocity of the particle.

- (a) (**1/2 point**) Write out the generalized momentum $\mathcal{P}_\mu = \partial\Lambda/\partial\dot{x}^\mu$ and the equations of motion in covariant form which correspond to the lagrangian.
- (b) (**1 point**) Write out the non-covariant lagrangian L which corresponds to the lagrangian (1). Simplify your result in the nonrelativistic limit.
Help: Pay attention that the non-covariant lagrangian L contributes to the action as $S = \int dt L$.
- (c) Assume that in the laboratory reference frame the tensor $S_{\mu\nu}$ in spherical coordinates is given by

$$S_{\mu\nu}^{(\text{spherical})} = \text{diag}\{S_{tt}, S_{rr}, S_{\theta\theta}, S_{\phi\phi}\} = \text{diag}\left\{\left(-\frac{a}{r}\right), -\frac{a}{r-a}, 0, 0\right\},$$

where a is some constant, and the particle always moves in the region $r \gg a$.

- i. (**1 point**) Identify all the continuous global symmetries of the action and construct the integrals of motion which may exist according to Noether's theorem.
- ii. (**1/2 point**) Analyze if Kepler's laws remain valid for this system. Can we consider that the particle's trajectory always remains in some plane in such a field?
- iii. (**1 point**) Integrate the equations of motion and find the trajectories, assuming that the particle moves in the plane $\theta = \pi/2$.